



MECH MAINT- BOILER Department
SIMHADRI

Note No :1



Ref:SMTTP/MECH MAINT-/2024-25/BMDMILLS/26762

Date:

09/04/2024(12:54:48)

Subject:Approval for Revision of LMI on Prevention Of Coal Pipe Choking & Procedure for Removal of Choke

- Local Management Instructions (LMI) on " **Prevention Of Coal Pipe Choking & Procedure for Removal of Choke** " at NTPC- Simhadri,has been reviewed and revised vide "LMI/OGN/OPS/MECH/045 REVISION NO.03, DATE: APR 2024". This revision is being initiated to incorporate the changes that has been brought due to revision of OGN "OGN/OPS/MECH/045 Rev. No.1, Oct 2021" vide OGN "OGN/OPS/MECH/045 Rev. No.02, Mar. 2024".
- The LMI has been revised as per new OGN "OGN/OPS/MECH/045 Rev. No.02, Mar. 2024" with following changes:
 - Clause 8.4 added "General Guidelines for prevention of Coal Pipe choking." against clause 5.4 of reference OGN.
 - Clause 9.5 Modified " to ensure that choking of PF pipe has been cleared " against clause 6.5 of reference OGN.
 - Clause 9.6 added "removal of temporary platforms" against clause Clause 6.6 of reference OGN.
- Submitted for kind approval of BUH please.

Submitted by:

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Note No:2

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Note No:3

The three changes incorporated in LMI at **clause 8.4 , 9.5 and 9.6** are detailed at page-1.

Reviewed and Submitted for approval please.

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On:10/04/2024(10:09:49) **Ref:**SMTTP/MECH MAINT-/2024-25/BMDMILLS/26762

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On:12/04/2024(09:27:54) **Ref:**SMTTPP/MECH MAINT-/2024-25/BMDMILLS/26762

Note No:6

Pl review the LMI and forward for further scrutiny/approval

Submitted by:

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Note No:7

LMI is reviewed and found in line with OGN. Submitted for approval please.

Submitted by:

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On:19/04/2024(15:47:29) **Ref:**SMTPP/MECH MAINT-/2024-25/BMDMILLS/26762

Note No:9

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On:19/04/2024(19:42:39) **Ref:**SMTPP/MECH MAINT-/2024-25/BMDMILLS/26762

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/MECH/045

REVISION NO. 3.00; DATE: Apr'2024

“PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE”

Approved for Implementation by

HOP, SIMHADRI_____

Date-----

LMI APPROVAL DOCUMENTS

1. TITLE OF LMI	PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE
2. a) DOCUMENT CODE b) REVISION/ISSUE NO.& DATE	LMI/OGN/OPS/MECH/045 REV: 3.00; APR'2024
3.ORIGINAL/REF.DOCUMENT	COS-ISO-00-OGN/OPS/MECH/045 Rev.No.02, March 2024
4.PREPARED BY: Sr. Manager (BMD/MILLS)	SHASHIKANT
5. CHECKED BY: DGM (MILLS) AGM (BMD) AGM (EEMG) GM (O) GM (O & M)	Shri. L R SWARNKAR Shri. OM PRAKASH Shri. UMA SHANKER VERMA Shri. T.R. RAJAGOPALAN Shri. SURESH JOHN DAVID
6. REVIEWED AND APPROVED BY	Shri. SANJAY KUMAR SINHA (HOP, SIMHADRI)
7. DATE OF ISSUE	
8.INDEX LIST	ENCLOSED.

NTPC	LMI	SIMHADRI
TITLE: PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE, REV: 3.00 DATE: APR'24		
REF NO. LMI/OGN/OPS/MECH/045		PAGE: 1 OF 8
<u>1.0 INDEX</u>		
<u>SECTION</u>	<u>TITLE</u>	<u>PAGE</u>
1.0	INDEX	1
2.0	INTRODUCTION	2
3.0	SCOPE	2
4.0	REFERENCE DOCUMENT	2
5.0	SUPERSEDED DOCUMENT	2
6.0	SPECIFIC INSTRUCTIONS	2
7.0	MECHANISM OF COAL PIPE CHOKING	2
8.0	PREVENTION OF COAL PIPE CHOKING	3
9.0	REMOVAL OF COAL PIPE CHOKING	4
10.0	REVIEW	5
11.0	REVISION INDEX	6
12.0	DISTRIBUTION LIST	7
13.0	COMPLIANCE RECORD	8

NTPC	LMI	SIMHADRI
TITLE: PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE, REV: 3.00		DATE: APR'24
REF NO. LMI/OGN/OPS/MECH/045		PAGE: 2 OF 8
<u>2.0 INTRODUCTION:</u> Coal pipe choking is a severe challenge faced in Pulverized coal power plant. The removal of choking is not only a rigorous and time taking job but hazardous and prone to fire and explosion. This document is prepared to ensure avoidance of Coal Pipe choking and safe procedure to be adopted to remove the choking if at all required. <u>3.0 SCOPE:</u> This LMI addresses two important steps to be followed: 1. Prevention of coal pipe choking by monitoring temperature of each coal pipe provided in control room. 2. Procedure for removal of coal from choked coal pipe, in case if there is an unavoidable choking. <u>4.0 REFERENCE DOCUMENT</u> COS-ISO-00-OGN/OPS/MECH/045 Rev.No.:02 March 2024 by operation services - NTPC Corporate Center. <u>5.0 SUPERSEDED DOCUMENT</u> COS-ISO-00-OGN/OPS/MECH/045 Rev.1, dtd. Oct 2021 and "LMI/OGN/OPS/MECH/045" Rev.2JAN'24. <u>6.0 SPECIFIC INSTRUCTIONS</u> NIL <u>7.0 MECHANISM OF COAL PIPE CHOKAGE</u> The inherent moisture in the coal is partially removed by the hot primary air inside the mill. Further, the pulverized fuel is discharged with a mill outlet temperature of 70⁰ – 75⁰ C. If this mill outlet temperature falls below 60⁰ C, there is a tendency of the pulverized fuel settling inside the PF pipes. Hence, it is important to monitor the mill outlet temperature and coal pipe temperature on a continuous basis. For this purpose, thermocouples have been provided on each and every coal pipes at critical locations both in stage I & II units. Other factors affecting the settlement of coal particles inside the PF pipe are the mill fineness and the PF velocity inside the coal pipe. Hence, it is important to maintain these parameters well within the limits. Responsibility: HOD-BMD & SCE-OPRN		

NTPC	LMI	SIMHADRI
TITLE: PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE, REV: 3.00		DATE: APR'24
REF NO. LMI/OGN/OPS/MECH/045		PAGE: 3 OF 8
8.0 <u>PREVENTION OF COAL PIPE CHOKING</u> 8.1 Monitoring of Coal Pipe temperature: Coal pipe temperature measurement RTD have been provided in all units & indication is available in control room. Coal pipe temperatures are being strictly monitored and maintained above 60degC during the whole shift and if there is a temp drop below 55 degree C in any coal pipe, then it is distinctively indicated in the control room in the LVS by red indication. At this point the coal feeding in the associated coal mill is to be reduced or feeder is to be taken out of service to ensure increase in temp. The trend of rise of coal pipe temperature is observed in comparison to other coal pipes. At the same time the coal pipe temperature is to be measured through temperature Gun at local. This is done to confirm that the feedback at CCR is not spurious. Also, the cap of sample point is to be opened to ensure whether coal/air mixture is coming out with pressure or not. Responsibility: SCE-OPRN 8.2 Steps to be followed in case coal pipe temp does not increase beyond 55° C. If there is no sufficient gain/rise in temperature of the pipe as compared to other 3 coal pipes, then mill is to be stopped and air is isolated. With the help of Mechanical maintenance, the rest three (3) MDVs (except the affected one) and burner isolation gates are to be closed. After that cold air is to be charged after keeping mill in service up to a maximum of 40 ton per hour for 2-3 hrs. sample point is opened to verify whether air is coming out with pressure or not, that is the confirmation that choking has been cleared. After this procedure MDV and burner isolation gates of rest 3 pipes are to be opened subsequently and hot air is to be charged to see whether the temperature of the concerned pipe rises. Responsibility: HOD-BMD & SCE-OPRN 8.3 Dirty pitot tube test Dirty pitot tube test to be done quarterly, to check any coal pipe velocity deviation that may lead to coal pipe choking and also immediately after clearing the coal pipe choke. Responsibility: HOD-EEMG & SCE-OPRN 8.4 General Guidelines for Prevention of Coal Pipe Choking (i)MDV, MPO including gaskets to be replaced in each OH / according to Guaranteed life. whichever is more or if punctured. (ii)Orifice inspection to be done in each Overhauling and replacement to be done if diameter. has increased by more than 1 %. (iii)Inspection of hanger & Support in each Unit Overhauling and rectification of defect if any. (iv)If deviation in PF velocity in each pipe comes more than 5 % in Dirty Pitot Test, then reasons for the same to be analyzed and corrective action (internal inspection of Mill, Inspection of venturi area, inspection of orifice and replacement if deviation found) should be taken at the earliest opportunity. After correction Dirty Pitot Test to be repeated. (v) Burner Inlet Gate (wherever provided) are to be kept operative and in healthy condition		

NTPC	LMI	SIMHADRI
TITLE: PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE, REV: 3.00		DATE: APR'24
REF NO. LMI/OGN/OPS/MECH/045		PAGE: 4 OF 8
9.0 <u>REMOVAL OF COAL PIPE CHOKING:</u> If after following the above procedure (8.1& 8.2), no improvement is observed, and persistence of the choke continues then following steps for “cutting window and subsequent air charging” are to be followed: 9.1 Mill cooling is done by charging cold air and bringing coal pipe temperature below 40 deg. MDV and burner isolation gate for the concerned pipe is to be closed. It is better, that the mill is further cooled to reach ambient temperature before start of work. The Permit to Work is to be processed and taken by maintenance Dept. The following isolations are to be followed during issuance of permit. • Mill to be isolated electrically. • HAG and CAG are to be closed and mechanically locked. Instrument air isolation to be done. • Burner isolation gate for concerned pipe is to be closed. • Seal Air • Inerting Steam The permit for working at height is to be taken prior to start of work. Hot work permit is also to be taken simultaneously. Safe approach and platform for reaching the height is to be built. The working area of the platform should be sufficiently large for comfortably cutting the window after maintaining a safe distance from the window. 9.2 Before cutting a window of required dimension, a small hole of 20mmX 20mm is made to ensure that there are no entrapped pressurized gases in the coal pipe. 9.3 The first window (approximately 200mm X 200mm) should always be cut on the top of the coal pipe i.e., on the plan view through gas cutting. The worker should keep his face away from the window to avoid any probable flame to reach his body. 9.4 After completion of window cutting, the cold air is to be charged to blow out and remove the chokages. If the removal of choke is achieved, then the process should be finished after closing the window by welding. Before charging of cold air adequate clearance to be logged in communication register in control room. 9.5 Even after this effort, if the removal of choke is not achieved then a second window would be cut in the same cross section of pipe at any of the side faces according to the site condition. Before starting of cutting the second window, a local cleaning of chocked coal should be done through poking from the already cut top window. The very purpose of such cleaning is to avoid any contact of flame of gas cutting set with the coal inside the pipe during cutting work. The process shall continue till the whole line is cleared which can be ensured from air coming out of sample point. If air is coming out from sampling point, window that was cut for cleaning to be welded back and a small new window 100 x 100 to be cut as near as possible to BIG (on MS portion and not on ceramic lined portion) in vertical pipe. Cold air is to be charged to see if air is coming. If air is coming then shut the air, take the PTW and window to be welded back. This activity to be carried out with BIG in closed condition wherever provided. While taking mill in service after return of PTW, temperature rise of PF pipe is to be checked to ensure that it is equal to other PF pipe before taking feeder in service. 9.6 Temporary platforms and all temporary supports to be removed after removal of choking. Responsibility: HOD-BMD & SCE-OPR		

NTPC	LMI	SIMHADRI
TITLE: PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE, REV: 3.00		DATE: APR'24
REF NO. LMI/OGN/OPS/MECH/045		PAGE: 5 OF 8
<u>10.0 REVIEW</u> This LMI shall be reviewed by the Head of O&M, on 2-yearly basis or as necessary.		

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NTPC	LMI	SIMHADRI
<u>13.0 COMPLIANCE RECORD</u>		
TITLE: PREVENTION OF COAL PIPE CHOKING AND PROCEDURE FOR REMOVAL OF CHOKE, REV: 3.00 DATE: APR'24		
REF NO. LMI/OGN/OPS/MECH/045		PAGE: 8 OF 8
UNIT NO.	ACTIVITY	COMPLIANCE